MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 12000015

(RATE SERIES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Rate Series Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-24308-1	Connectors, Electric, Rectangular, Miniature Polarized Shell, Rack and Panel General Specification for
MIL-C-50785	Circuit Card Assembly, Sight Current Generator, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

12000239	Sight Current Generator
12000015	Circuit Card Assembly, Rate Series
12000091	Microcircuit, Multiply/Divide

PACKAGING DATA SHEET

12000015	Packaging of Circuit Card Assembly, Rate Series

(Copies of specifications, standards, drawings, and publications required by Supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1285

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3. REQUIREMENTS

3.1 Item definition.- This assembly is a single-ended printed circuit board consisting of five operational amplifiers and three divider modules. The assembly is used to generate the rate series input term of the magnet current and torque current voltage circuitry, when installed in the Sight Current Generator Drawing Number 12000239.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 12000015.

3.1.2 Government furnished property list.- Not applicable.

3.2 Characteristics.

3.2.1 Performance.- The assembly shall be capable of performing requirements a through d when assembled in accordance with Drawing Number 12000015.

a. Operational amplifiers AR1 and AR2 shall sum and invert the input signals that make up the super elevation term of the torque current voltage circuit.

b. Operational amplifiers AR3 and AR4 in conjunction with transistors Q1 and Q2 shall amplify and scale the range rate input voltage.

c. Dividers U1 and U2 in conjunction with operational amplifiers AR3 and AR4 shall generate the rate series term of the magnet current voltage circuit.

d. Divider U3 and operational amplifier AR5 shall generate the lead angle term of the magnet current voltage circuit.

3.2.1.1 Input and output levels.

3.2.1.1.1 Input power.- Input power shall be supplied to the assembly as specified in Table I.

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TABLE I - INPUT VOLTAGES

PIN	VOLTAGE
14	+15 Vdc ± 2.0 percent
2	-15 Vdc ± 2.0 percent
1	Ground
18	+7 Vdc ± 0.1 percent
3 or 16	Triangle wave -8.47 ± 1.03 Vdc to +8.47 ± 1.03 Vdc within a period of 1.00 millisecond ± 10 percent.
4 or 17	Triangle wave -8.47 ± 1.03 Vdc to + 8.47 ± 1.03 Vdc within a period of .8 millisecond ± 10 percent.
13	+7 Vdc ± 0.1 percent

3.2.1.1.2 Output levels.- The requirements specified in 3.2.1.1.2.1 through 3.2.1.1.2.4 shall be met when supplied with the specified voltage input (Vin). The inputs applied to the microcircuits U1, U2 and U3 shall not exceed the electrical requirements specified on Drawing Number 12000091.

3.2.1.1.2.1 Pin 7 output.- The output between pin 7 and pin one shall be $\frac{V_{38} \times 0.1\ V_4}{-V_{16}} \pm 2.0$ percent ± 0.021 Vdc, with a Vin applied between pin 19 and pin one.

V_{38} = (0.5385) Vin

V_4 = +7 Vdc ± 0.1 percent

V_{16} = +0.7 to +7 Vdc

Vin = 0 Vdc to +13 Vdc

3.2.1.1.2.2 Pin 10 output.- The output between pin 10 and pin one shall be $\frac{0.7\ V_{38}}{-V_{16}}$ ± 2 percent ± 0.021 Vdc, with a Vin applied between pin 15 and pin one and +15 Vdc applied between pin 9 and pin one.

V_{38} = (0.7) Vin

V_{16} = +0.7 to +7 Vdc ± 0.1 percent

Vin = -10 Vdc to +10 Vdc

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3.2.1.1.2.3 Pin 11 output.- The output between pin 11 and pin one shall be $\frac{V_{38} \times 0.1\ V_4}{-V_{16}} \pm 4.0$ percent $\pm$ .030 Vdc, with a Vin applied between pin 15 and pin one and +15 Vdc applied between pin 9 and pin one.

V_{38} = output of pin 10

V_{16} = +0.7 to +7 Vdc $\pm$ 1.0 percent

V_4 = +7 Vdc $\pm$ 0.1 percent

Vin = -10 Vdc to +10 Vdc

3.2.1.1.2.4 Pin 12 output.- The output between pin 12 and pin one shall be as specified in a through d.

a. Zero $\pm$ .012 Vdc with zero Vdc applied between pins 15, 21, 22 and pin one, and +15 Vdc applied between pin 9 and pin one.

b. Vin (-0.185) $\pm$ 2.0 percent with Vin of zero to +7 Vdc applied between pin 21 and pin one, zero Vdc applied between pins 15, 22 and pin one, and +15 Vdc applied between pin 9 and pin one.

c. Vin (-0.216) $\pm$ 2.0 percent where Vin is the pin 25 output, zero Vdc is applied between pins 15, 21 and pin one, and +15 Vdc applied between pin 9 and pin one.

d. Vin (-0.054) $\pm$ 2.0 percent where Vin is the pin 6 output, zero Vdc is applied between pins 15, 21 and pin one, and -15 Vdc applied between pin 9 and pin one.

3.2.1.2 Fault isolation test points.- The assembly has the following test points available to facilitate fault isolation pin 5, 6, 8, 20, 25, TPI-5. These points are not output pins.

3.2.1.3 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristics.- The physical characteristics shall be as specified in drawing 12000015 and the engineering parts list.

3.3 Design and construction.

3.3.1 Production drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 12000015 and associated lists.

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3.3.2 Standards of manufacture.- The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-50785.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment.

4.2.1.1 Power supply (PS1).- The power supply shall be 15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 Power supply (PS2).- The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.3 Power supply (PS3).- The power supply shall be 7.0 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) or 100 mv peak-to-peak.

4.2.1.4 Power supply Ø1 (PS4).- The power supply shall generate a triangle wave -8.47 ± 1.03 Vdc to +8.47 ± 1.03 Vdc within a period of 1.00 millisecond ± 10 percent.

4.2.1.5 Power supply Ø2 (PS5).- The power supply shall generate a triangle wave -8.47 ± 1.03 Vdc to +8.47 ± 1.03 Vdc within a period of 0.8 millisecond ± 10 percent.

4.2.1.6 Power supply (PS6).- The power supply shall be zero ± 13 Vdc, with accuracy of adjustment and regulation to ± .01 percent, 10 ma or greater.

4.2.1.7 Power supply (PS7).- The power supply shall be zero ± 13 Vdc, with accuracy of adjustment and regulation to ± .01 percent, 10 ma or greater.

4.2.1.8 Digital voltmeter (DVM).- The digital voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

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4.2.1.9 Switch (S1).- One double pole single throw switch is required.

4.2.2 Functional test.

4.2.2.1 Test connections.- Power supplies PS1 thru PS7 shall be connected to the assembly under test in accordance with Figure 1 using receptacle M24308/1-3 of specification MIL-C-24308-1.

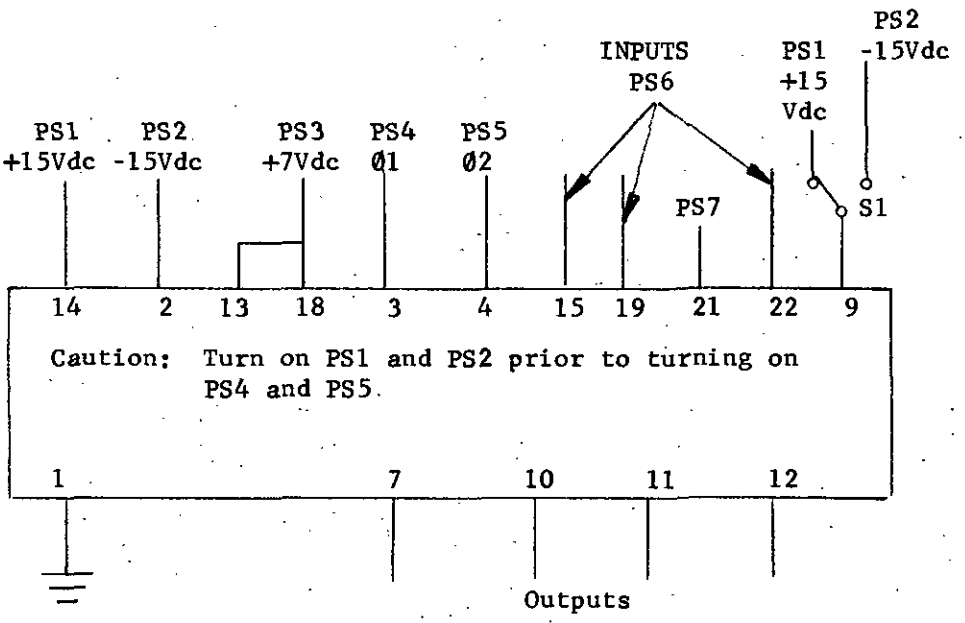


Figure 1. Test Connections

4.2.2.2 Test procedure.- Compliance with the requirements of 3.2.1.1.2 through 3.2.1.1.2.4 shall be verified as follows:

a. Connect PS6 to pin 19, connect PS7 to pin 21. Adjust power supplies PS6 and PS7 as specified in Table II. Set S1 to +15 Vdc.

TABLE II - CONNECTIONS

Pin 19 Input Vdc	Pin 21 Input Vdc	Pin 7 Output Vdc
+13 ± 0.007	+7 ± 0.007	-0.7 ± 0.035
0 ± 0.001	+7 ± 0.007	0 ± 0.030
+13 ± 0.007	+0.7 ± 0.007	-7 ± 0.160

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b. Connect PS6 to pin 15, connect PS7 to pin 21 adjust power supplies PS6 and PS7 as specified in Table III. Set S1 to +15 Vdc.

TABLE III - CONNECTIONS

Pin 15 Input Vdc	Pin 21 Input Vdc	Pin 10 Output Vdc
+10 ± 0.007	+7 ± 0.007	-0.7 ± 0.035
0 ± 0.001	+7 ± 0.007	0 ± 0.030
-10 ± 0.007	+0.7 ± 0.007	+7 ± 0.160

c. Connect PS6 to pin 15, connect PS7 to pin 21 adjust power supplies PS6 and PS7 as specified in Table IV. Set S1 to -15 Vdc.

TABLE IV - CONNECTIONS

Pin 15 Input Vdc	Pin 21 Input Vdc	Pin 10 Output Vdc
+7.5 ± 0.007	+7 ± 0.007	0 ± 0.030
0 ± 0.001	+0.7 ± 0.007	+7 ± 0.160

d. Connect PS6 to pin 15, connect PS7 to pin 21. Adjust power supplies PS6 and PS7 as specified in Table IV. Set S1 to -15 Vdc.

TABLE V - CONNECTIONS

Pin 15 Input Vdc	Pin 21 Input Vdc	Pin 11 Output Vdc
+10 ± 0.007	+7 ± 0.007	0 ± 0.035
0 ± 0.001	+7 ± 0.007	-0.070 ± 0.035

e. Connect PS6 to pin 15, adjust PS6 to +10 ± 0.007 Vdc set S1 to +15 Vdc. The output measured at pin 12 with the DVM shall be -0.374 ± 0.020 Vdc.

f. Connect PS7 to pin 21, adjust PS6 to +7 ± 0.007 Vdc set S1 to +15 Vdc. The output measured at pin 12 with the DVM shall be -1.296 ± 0.030 Vdc.

g. Connect PS6 to pin 22, adjust PS6 to -15 ± 0.007 set S1 to +15 Vdc. The output measured at pin 12 with the DVM shall be -1.512 ± 0.035 Vdc.

h. Set S1 to +15 Vdc. The output measured at pin 12 with the DVM shall be 0 ± 0.012 Vdc.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC Form 1029, 12000015.

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6. NOTES.

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

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Preparing Activity:

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Project No. 1285-A434

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE: MIL-C-50785/15(MU) Circuit Card Assembly: 12000015 (Rate Series)

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	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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